

Gaoxiang Zhao

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Education

University of Pennsylvania

August 2025 – May 2027

MS in Robotics

Philadelphia, USA

- Related Courses: Machine Learning, Learning in Robotics, Physical Intelligence

Wuhan University

September 2021 – July 2025

BEng in Communication Engineering

Wuhan, China

Experience

University of Pennsylvania

December 2025 – Present

Research Intern (Advisor: Kostas Daniilidis)

Philadelphia, USA

- Investigating event-based vision algorithms for precise Hand-Object Pose Estimation in high-speed robotic manipulation tasks.

Manycore Tech

August 2024 – November 2024

Research Intern

Hangzhou, China

- Developed Monte-Carlo physically-based rendering algorithms for a production-scale rendering pipeline supporting photorealistic synthetic data generation.

Projects

Sim2Real Drone Racing with Deep RL

February 2026 – April 2026

- Designed a PPO-based RL pipeline in Isaac Lab with 8,192 parallel environments to train a Crazyflie 2.1 for autonomous 3-lap, 7-gate racing, using a 36D body-frame observation (velocity, rotation matrix, gate corners) and a 10-term reward function.
- Achieved zero-shot sim-to-real transfer via dynamics domain randomization (thrust-to-weight, drag, PID gains) and three iteration-conditioned curricula (approach distance, DR magnitude, speed reward weighting), completing 3 laps in 20s on a real Crazyflie 2.1.

Forward and Inverse Kinematics Solver

September 2025 – November 2025

- Developed a C++ kinematics toolkit with forward/inverse kinematics and quaternion-based orientation interpolation.
- Implemented three IK solvers (Limb-based, CCD, Pseudo-Inverse Jacobian).

Physically-based Renderer

December 2023 – January 2025

- Developed a physically-based rendering engine in C++, using Monte-Carlo integration and multiple importance sampling for accurate light transport simulation.
- Implemented Path Tracing, Bidirectional Path Tracing, and photon mapping for physically accurate global illumination in complex scenes.

Diffraction Simulation

June 2024 – August 2024

- Implemented the Free-Space Diffraction BSDF (Steinberg et al., 2024) in the PBRT-v3 renderer, extending Monte Carlo path tracing with edge-based Fraunhofer diffraction.
- Added continuous spectral rendering on top of the original monochromatic and RGB tricolor modes to study wavelength discretization effects on diffraction patterns.

Technical Skills

- Programming: C++, Python, MATLAB
- Tools: $\text{\LaTeX}$, Linux, Git, Slurm, PyTorch, ROS, Unity, OpenGL, Isaac Lab, MuJoCo
- Language: English (Proficient), Mandarin (Native)